National University of Computer & Emerging Sciences
Karachi Campus



AS-7: AegisSwarm-7

Autonomous Drone Swarm Defense System

Project Proposal
Object Oriented Programming
Section: BCS-2E

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Introduction

AS-7, short for AegisSwarm-7, is a program designed to enhance air defense systems through a swarm of autonomous drones. Unlike traditional defense systems, which are stationary, AS-7 focuses on a many-to-many attack strategy based on an array of low-cost interceptor drones, executed with the help of Object Oriented Programming concepts like Inheritance, Aggregation and Composition. The program is also designed to protect critical infrastructure including schools, health facilities and power plants. AS-7 serves as an efficient yet economically friendly “shield”, protecting civilians, strategic infrastructure, and country’s defense expenditure.

Existing System

Currently, Pakistan uses **HQ-9P** for long distance air defense. The HQ-9P is an air defense system primarily for intercepting high priority targets, and comes with high mobility making transportation easier. The system uses the HT-233 Radar, which has a range of 50 km, and can detect up to 100 targets.

The Iranian air defense mainly uses **Bavar-373** for its long distance targets. It is similar to the Russian S-400 and can track up to 300 targets using Sayyad-4 missiles. High mobility is ensured through the Zoljanah 10x10 truck.

Problem Statement

Current air defense systems, like HQ-9P (Pakistan) and Bavar-373 (Iran), are designed to target jets and missiles, making them powerful but extremely expensive. The problem which arises is that while the radar can detect into a wide range, these missile-focused systems can only engage around 6-8 targets simultaneously. Now if an enemy sends 30 low-cost drones, these systems will be overwhelmed and would not be able to shoot all 30 down. This would also be inefficient since these expensive missiles would be used to shoot down 30 cheap drones.

Proposed Solution

AS-7 is designed to counter enemy attacks through the use of a swarm of low-cost autonomous drones. AS-7 focuses on being budget friendly as well as being effective. While defense systems like HQ-9P can only engage 6-8 targets simultaneously, each AS-7 drone acts as a sensor which comes with real time communication between the decentralized swarm, widening its radar for detection as well as engagement. The main purpose of the system is to act as a "Digital Shield" against enemy threats, ensuring public safety, protecting strategic infrastructure, and maintaining economic balance.

Salient Features

Decentralized Swarm Logic: Each drone communicates its coordinates and status with the rest of the swarm, ensuring that the system works even if one of the drones goes down.

Location Shield: The system is designed to protect important infrastructure including hospitals, schools and power plants. Each drone checks the status of its final attack coordinates and decides whether to use a different point according to the location.

Target Prioritization: The system decides how many drones are supposed to be dispatched according to threat type (e.g. small loitering drones, high speed missiles).

Coordinate Calculation: Instead of using heavy warfare, the system uses precise math to calculate the exact set of coordinates for the target's trajectory making it efficient.

Automatic Handover Logic: Each drone keeps track of its battery, and updates the swarm and base about its status, so that when a drone's battery is low, another drone is deployed for replacement while the original returns to base.

Tools and Technologies

Programming Language: C++ (OOP)
Framework: Standard Template Library (STL)
Development Environment: Dev C++, VS Code, Programiz, Github
Operating System: Windows 11